Université Marie & Louis Pasteur

Internship Report

Master in Competitive and Business Intelligence

Master BDEEM

Behavioral and Digital Economics for Effective Management

Edge-Based Defense in Networks

A Modification of Bloch, Chatterjee & Dutta (2023)

with Applications to Sustainable Resource Management

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Abstract

This report studies what happens when defense in a network attack game is moved from the nodes to the connections between them. The starting point is the model by Bloch, Chatterjee and Dutta (2023), where each target invests in protection at its own location and an attacker chooses both a target and a path to reach it. The idea of placing defense on connections instead of locations comes from Mavronicolas et al. (2008), who showed that this small change produces equilibria with very different structure.

Keeping Bloch's setup of continuous investment and independent defenders, the report builds the modified model in full, derives the main results, and explains the reasoning behind each finding. The conclusions are illustrated using networks that resemble real systems — power grids, water distribution networks, and shared environmental infrastructure. An interactive R Shiny simulator was built to test the results numerically and to allow further experiments. The work connects two strands of the literature that have so far stayed apart: continuous-investment games on networks, and edge-based defense.

Contents

General Introduction

The Setting

Many of the systems we depend on every day are networks. Electricity travels through a grid of transmission lines connecting power stations to substations to households. Water moves through a system of pipes and pumping stations. Goods travel along supply routes connecting producers to warehouses to shops. These networks are built to deliver something useful, but they share one uncomfortable feature: they can be attacked.

An attack on a network does not always mean a deliberate one. A storm can take down a transmission line. A pipe can rupture. A road can flood. Whether the cause is deliberate or accidental, the result is the same — damage moves through the network along whatever connections it can find. Protecting these systems matters, and it matters more as we ask them to do more. The push for sustainable resource management makes this question sharper. We are connecting power grids across borders, sharing water across regions, and building supply chains that span continents. The benefits are real, but so is the new exposure that comes with all that connection.

The Question

How should defense be organized in such a network? Two basic answers exist in the academic literature. The first places defense on the nodes themselves — on the power stations, the pumping stations, the warehouses. Each location takes care of itself, and an attacker who reaches a defended location is intercepted there. The second places defense on the connections — on the transmission lines, the pipes, the roads. The same connection can be watched from either end, and an attacker trying to use it must get past whatever defense both ends have put in place.

These two arrangements lead to very different strategic situations. In the first, each location decides on its own how much to defend. In the second, the two ends of any connection share something — and that sharing changes how each side thinks about its own choices. Do they cooperate? Do they free-ride on each other? Does it matter how many connections a location has?

This report studies the second arrangement. It builds on a recent model by Bloch, Chatterjee and Dutta (2023), which places defense on nodes, and asks what changes when defense is moved to connections instead. The inspiration for the move comes from Mavronicolas et al. (2008), who studied a related game where the defender protects connections directly and showed that doing so produces a recognizably different equilibrium structure.

What This Internship Did

The internship took place at CRESE, the economic research center at Université Marie et Louis Pasteur in Besançon, under the supervision of Professor Sylvain Béal. The work had three connected goals.

The first goal was to build the modified model in full. This meant deciding exactly how to translate every piece of Bloch's original formulation into the new setting where defense lives on connections. Some pieces carry over without change. Others change in ways that are obvious. A few change in ways that are not obvious at all, and the report works through each of these carefully.

The second goal was to derive the main theoretical results. The model needed to be shown to have an equilibrium, the equilibrium needed to be shown to be unique, and the conditions defining the equilibrium needed to be written out explicitly. From there, the report examines how the equilibrium responds to changes in the network: a target becoming more valuable, a new connection being added, or a target being removed from consideration. Each of these results is presented with the reasoning behind it explained as plainly as possible.

The third goal was to make the results accessible through a working tool. An R Shiny simulator was built that takes any network the user describes, computes the equilibrium under both defense models (Bloch's original node defense and the modified edge defense), and presents the comparison side by side. The simulator served two purposes: it allowed the theoretical results to be verified numerically on real examples, and it allowed comparative experiments to be run in seconds rather than hours.

The Central Problem

The central question this report addresses is the following: *when defense in a network is placed on the connections rather than the locations, how does the strategic situation change, and what does this tell us about the protection of distributed infrastructure?* The question is theoretical in form but practical in motivation. Real infrastructure — power lines, pipelines, transport routes — is defended at least partly along its connections, not only at its endpoints. Understanding how this changes the strategic balance between attackers and defenders is the contribution this work tries to make.

Structure of the Report

The report is organized into three chapters. Chapter 1 presents the research landscape: the broader literature on network security games, the specific tradition that Bloch's model belongs to, and the smaller body of work on edge-based defense. The chapter ends by identifying the gap that this work fills.

Chapter 2 develops the modified model. It begins with the practical motivation, sets up the notation, presents the differences from Bloch carefully, and works through the formal results. Numerical examples are used throughout to illustrate why each result holds. Two further questions raised during the supervision are also addressed: what happens if the cost of defense on a shared connection is paid jointly rather than independently, and what role information plays when defenders only know about their immediate neighbors.

Chapter 3 covers the implementation. The R Shiny simulator is described, the numerical results are reported, and the chapter closes with a reflection on the skills developed during the internship and the lessons learned.

The general conclusion summarizes the findings, acknowledges the limits of the work, and suggests directions for future research.

Chapter 1

The Research Landscape

Introduction to the Chapter

This chapter places the report's contribution within the broader academic conversation about how networks should be defended against attacks. The literature on this question has grown in two distinct directions, which until now have stayed largely separate. One direction looks at how defenders allocate resources at the locations of a network. The other looks at how defenders interfere with the connections between locations. The model studied in this report sits between these two traditions: it borrows the continuous-investment framework from the first and the placement of defense from the second. To make the contribution clear, the chapter walks through both traditions, identifies what each one offers and what each one leaves out, and then states explicitly the gap that this work fills.

1.1 Network Security Games: The Field

A network security game is a simplified picture of a real situation in which one or more attackers try to do harm to a system represented as a network, and one or more defenders try to prevent that harm. The network is usually drawn as a set of points (called nodes) connected by lines (called edges). The attacker chooses a target — typically one of the nodes — and tries to reach it. The defender allocates protective resources somewhere in the network, hoping to stop the attack before it succeeds.

The reason such a simplified picture is worth studying is that it captures something real. Power grids, water distribution networks, communication systems, supply chains, and even ecological corridors all have the property that they are made of locations linked by connections, that they carry something valuable, and that they are exposed to disruption. The strategic question — how to protect a network when its defenders have limited resources and its attackers are intelligent — is the same across all of these settings, even when the technical details differ.

The field has produced a wide range of models, distinguished from one another by three main choices. First, who is the defender: a single central authority deciding for the whole network, or many independent defenders each protecting only their own piece? Second, what is the form of defense: an all-or-nothing decision to protect a location, or a continuous investment that gradually raises the chance of interception? Third, where is defense placed: at the nodes themselves, or on the connections between them?

Different combinations of these three choices produce different models. The work in this report can be located by the choices it makes: the defenders are independent (decentralized), their investment is continuous, and their defense is placed on the connections. This combination is, to my knowledge, new. The chapter that follows shows what already exists in the surrounding literature.

1.2 Defense at the Nodes: The Centralized Tradition

A first body of work looks at defense placed at the nodes of a network, with a single centralized defender deciding on the whole protection plan. The reasoning is straightforward: the defender owns or coordinates the network, can see it as a whole, and chooses where to put protection so as to minimize the damage from a future attack.

Goyal and Vigier (2014) study a version of this problem where the attack spreads from an initial node to its neighbors, like a contagion. The designer of the network chooses which connections to build and how to allocate defense at the nodes; the attacker, moving second, chooses where to strike. The main message of the paper is that the most effective network shapes are surprisingly concentrated — the defender is often best off building a star network and protecting the center, even if doing so leaves many nodes exposed. Defense at the center stops the contagion before it can spread.

Dziubiński and Goyal (2013) continue this line by allowing the designer to choose both the network and the defense at the same time, against an attacker who is given a fixed budget for striking nodes. Their results show that the optimal defense is sharply binary: either the network has zero defense, exactly one protected node, or every node protected. Intermediate solutions are rarely optimal. The paper produces a clear taxonomy of when each regime is best, depending on the cost of building connections and the cost of defending nodes.

Dziubiński and Goyal (2017) sharpen the analysis further by introducing the idea of a network separator — a set of nodes whose removal disconnects the network. The defender's problem becomes the problem of choosing which separator to protect. This more abstract reformulation produces some surprising results, including a version of Braess's paradox: adding a connection to a network can sometimes hurt the defender, because it creates a path that the attacker can exploit.

Cerdeiro et al. (2017) change one important assumption: instead of a centralized defender, each node decides on its own whether to protect itself. The contagion structure is similar to Goyal and Vigier (2014), but the strategic situation is different because individual nodes do not internalize the benefit their protection gives to others. The paper measures the cost of decentralization and shows that, in many cases, it is severe — decentralized choice can lead to dramatic underinvestment in protection compared to the centralized optimum.

What unites these papers is that defense is placed on nodes, and is typically all-or-nothing. The strength of the tradition is that it produces clean, sharp characterizations of optimal network design and protection. Its limitation is that real defense is often not all-or-nothing — it is a question of degree — and is often paid for not at locations but along connections.

1.3 Continuous Defense at the Nodes: Bloch, Chatterjee & Dutta (2023)

Bloch et al. (2023) occupies a particular position in this literature. Like the papers above, defense is placed at the nodes. Unlike them, defense is continuous: each defender chooses how much to invest, with a quadratic cost. And unlike the centralized tradition, the defenders are independent — each target node makes its own investment decision based on its own value and its own perception of the threat.

The model has several distinctive features. The attacker chooses both a target and a path to reach it, and so the attack itself moves through the network rather than landing instantly. Each node along the path may intercept the attack, with a probability that depends on its own investment. Targets have heterogeneous values, and the values are private in the sense that each defender only cares about protecting its own. The result is a richer game than the binary models above: the equilibrium involves not only which targets are attacked but also how much each defender invests, how the attacker spreads probability across paths, and how the network's topology shapes both decisions.

Bloch and his coauthors show that under mild conditions the game has a unique equilibrium. They characterize that equilibrium fully for several network shapes, including line networks, trees, and stars. They derive comparative statics: how the equilibrium shifts when a target's value changes, when a new connection is added, or when a target is dropped from the attacker's consideration. They also study a cooperative version of the model, in which a single coordinator decides defense for all nodes. The cooperative arrangement makes the attacker strictly worse off than under decentralized defense, because the coordinator can engineer changes to the support that no individual defender would choose alone.

The model in Bloch et al. (2023) is the foundation on which this report builds. Its strength is the combination of continuous investment, decentralized choice, and a path-based attacker — features that together make the model both realistic and tractable. Its limitation, from the point of view of this report, is that defense remains located at the nodes. The question this report asks is what happens if that one feature is changed.

1.4 Defense at the Connections: The Edge Tradition

A second body of work, much smaller, places defense on the connections themselves. In these models, the defender does not protect individual nodes; instead, the defender controls or interferes with the edges through which an attacker must move.

Washburn and Wood (1995) present the earliest version of this idea. A single attacker tries to move through a network from a source to a destination; a single defender chooses one edge to inspect, hoping to catch the attacker on it. The game is repeated daily, and both players use mixed strategies. The authors show that the value of the game can be computed using the maximum-flow / minimum-cut structure of the network — a connection that turns out to be central to the entire literature on edge-based interdiction.

Israeli and Wood (2002) extend the same idea to a richer setting in which the defender can interdict multiple edges, subject to a budget. Each interdicted edge becomes longer or impassable, and the attacker — the follower in this version of the game — chooses the shortest path that remains. The paper develops an algorithmic approach (using a technique called covering decomposition) that performs much better than direct optimization. The structural intuition is striking: when the attacker has many alternative paths, the defender must block enough of them to eliminate all the cheap routes. This idea — that the defender's job is to "cover" all the paths the attacker could take — reappears in the edge-based modification studied in this report.

Zenklusen (2010) approaches the same question from a different angle, studying the computational complexity of edge interdiction on planar graphs — networks that can be drawn without crossings, like most real geographical networks. The paper develops algorithms that are tractable for these special cases and shows their limits for general networks. While not directly used in this report, the work is part of the same conceptual tradition: defense lives on edges, and the question is how to allocate it efficiently.

Mavronicolas et al. (2008) stands somewhat apart. The setup is closer in spirit to the present work, even though the technical details differ. Multiple attackers each choose a node, and a single defender chooses one edge to protect; defending an edge covers both of its endpoints simultaneously. The authors show that under suitable conditions the equilibrium has an elegant matching structure: the attacked nodes form an independent set (none of them adjacent), and the defender's chosen edges form a matching that pairs each non-attacked node to a unique attacked one. This conceptual idea — that defense on an edge produces coverage spillover between its endpoints — is the direct inspiration for the modification studied in this report.

The strength of the edge tradition is that it captures something the node tradition misses: defense is often paid for along connections, not at locations, and shared connections create strategic interactions that node-based models cannot represent. The limitation is that the edge tradition has so far worked mostly with discrete, all-or-nothing defense by a centralized defender. The combination of continuous investment, decentralized defense, and edge placement has not been studied.

1.5 The Gap

Two strands of literature, each with its own strengths and limitations, have grown in parallel. The node tradition has produced the most refined models of decentralized continuous defense, with Bloch et al. (2023) as its most developed example. The edge tradition has produced powerful results about the structure of optimal interdiction, but has stayed within discrete, centralized formulations.

What is missing is a model that combines the realism of continuous decentralized investment with the structural insight of edge-based defense. The present work is an attempt to fill this gap. It takes Bloch's framework as its starting point and modifies one feature: defense is moved from the nodes to the edges, with both endpoints of an edge able to invest independently. The modification is small to state but consequential to derive. It changes how survival probabilities are computed, how defenders interact with one another, how multi-path targets behave, and how cooperative defense compares to decentralized defense. Each of these consequences is the subject of the next chapter.

Conclusion to the Chapter

The literature on network security games has converged on a common set of questions but has answered them in two different ways. Defense at the nodes offers tractability and realism for individual locations; defense at the edges offers a clearer picture of how connections matter. Bloch et al. (2023) brings the node tradition to a high level of sophistication. Mavronicolas et al. (2008) shows what edge-based defense can look like in a discrete world. The present work bridges the two by asking what happens when continuous, decentralized defense is placed on edges — and, as the next chapter will show, the answer is not a small adjustment but a meaningful change in the strategic landscape.

Chapter 2

The Edge-Based Defense Model

Introduction to the Chapter

This chapter develops the contribution of the report. It begins with the practical motivation that justifies moving defense from locations to connections, drawing on examples from sustainable resource management. It then summarizes the original model of Bloch, Chatterjee and Dutta (2023) so the reader has a clear reference point. The modified model is built up step by step, with every piece of notation introduced once and used freely afterward. After that, the chapter walks through the similarities with Bloch's model and then the differences, explaining the reasoning behind each change as plainly as possible. Two worked examples illustrate the main effects. The chapter closes with two further questions raised during the supervision: what happens if the cost of defense on a shared connection is paid jointly, and what role information plays when defenders only know about their immediate neighbors.

The mathematical content of this chapter is presented through formal statements followed by plain-language explanations and short proof sketches. The full proofs are collected in the Appendix, where any reader who wants to verify a result step by step can find it.

2.1 The Practical Motivation

The networks that matter most for sustainable resource management are not abstract objects. They are the physical and organizational systems through which we move energy, water, materials, and information across regions and across borders. When we ask how such systems should be defended, the question is not only about who pays for protection but about *where* protection is placed.

Consider an electricity grid. Transmission lines connect generating stations to substations and substations to households. A line is a long, exposed object running across the landscape, vulnerable to weather, vegetation, accidents, and deliberate attack. The substations at either end of the line both have an interest in keeping the line intact: each loses something if the line goes down. Each can install monitoring equipment, send patrols along the line, or invest in surge protection. The defense of the line is shared between its two endpoints in a way that the defense of a single substation is not.

Water distribution networks have the same structure. A pipe runs from a treatment facility to a distribution hub, or from one regional reservoir to another. Both ends benefit from the pipe staying intact and uncontaminated. Both can install sensors, schedule inspections, or fund maintenance. A single pipe is defended jointly by the two facilities it connects, even when those facilities are managed by different organizations.

Cross-border environmental infrastructure follows the same logic. A river that crosses two countries carries pollution, water rights, and ecological consequences in both directions. Both countries have an interest in monitoring the river, and both can invest in monitoring stations along it. The river itself — the connection — is what is being defended, not the individual cities or facilities at either end of it.

These three examples share a feature that the original Bloch model does not capture. Defense in real systems is often paid for along connections, not at locations, and the two ends of any connection share both the threat and the cost. The strategic situation that arises from this sharing — whether the two ends cooperate, whether they free-ride on each other, whether the structure of the network amplifies or dampens these effects — is the question the rest of this chapter addresses.

2.2 Bloch's Original Model

Before introducing the modification, we summarize the model of Bloch, Chatterjee and Dutta (2023). The reader who is already familiar with the paper can skim this section; for everyone else, it provides the reference point against which the modified model is built.

The model is played on a network $G = (V, E)$ where $V = \{0, 1, 2, \ldots, n\}$ is a set of nodes and E is a set of directed edges. Node 0 is the attacker. Nodes $1, \ldots, n$ are potential targets. Each target i has a value $b_i \in (0, 1)$, representing the damage caused if a successful attack reaches it. Values are assumed generic, meaning $b_i \neq b_j$ for all $i \neq j$.

The attacker chooses a probability distribution π over the set of paths $\mathcal{P}$ from node 0 to target nodes. Each path p terminates at some target $k(p)$. By summing the probabilities of all paths ending at the same target, the distribution over paths π induces a distribution q over targets: $q_i = \sum_{p:k(p)=i} \pi(p)$.

Each target i chooses a single interception investment $x_i \in [0, 1]$. This investment has a cost of $x_i^2/2$. When an attacker passes through node i on the way to its destination, the probability that node i intercepts the attacker is x_i. Equivalently, the probability that the attacker survives passage through node i is $(1 - x_i)$. The investment x_i applies regardless of which direction the attacker comes from — this is the central feature of node defense.

The probability that the attack survives an entire path p to reach the target $k(p)$ is the product of survival probabilities along the path:

$$\alpha_{k(p)}(p) = \prod_{\substack{j \in V(p) \\ j \neq 0,\, j \neq k(p)}} (1 - x_j),$$

where $V(p)$ is the set of nodes on path p. The probability of a successful attack on $k(p)$ is then $\beta_{k(p)}(p) = \alpha_{k(p)}(p) \cdot (1 - x_{k(p)})$, with the additional factor accounting for interception at the target itself.

The attacker's expected payoff is:

$$U(\pi, x) = \sum_{p \in \mathcal{P}} \pi(p) \cdot \beta_{k(p)}(p) \cdot b_{k(p)}.$$

Each defender i has the payoff:

$$V_i(\pi, x) = -\sum_{\substack{p \in \mathcal{P} \\ k(p) = i}} \pi(p) \cdot \beta_i(p) \cdot b_i \; - \; \frac{x_i^2}{2}.$$

The first term is the expected loss from attacks targeting node i; the second is the cost of investment.

Bloch and his coauthors prove existence and uniqueness of the Nash equilibrium under the genericity assumption. They derive equilibrium investments and attack probabilities for first targets (those reachable directly from the attacker through unprotected territory) and for deeper targets (those reached through other support members). They also study a cooperative version of the game where a single coordinator chooses defense for all nodes.

The defining feature of the model, for the purposes of this chapter, is that defense is placed at nodes. A single investment x_i defends the node from *any* direction of approach. The next section moves this defense to the edges and works out what changes.

2.3 The Edge-Based Modification

We now describe the modified model. The network, the attacker, the target values, and the genericity assumption are all retained from Bloch's formulation. What changes is how defenders invest and how interception occurs.

[Edge investment] Each defender $i \in \{1, \ldots, n\}$ chooses a vector of investments $(x_{i,e})_{e \in E(i)}$, where $E(i) = \{e \in E : i \in e\}$ denotes the set of edges incident to node i. Each investment $x_{i,e} \in [0, 1]$ is chosen independently and represents the interception effort that node i places on edge e.

In words, each node is no longer choosing a single defense level; instead, it chooses a separate level for each connection it participates in. This models decentralized self-defense: a node allocates protective resources across the connections it shares with others, possibly investing more on connections it perceives as more vulnerable.

[Edge survival probability] For an edge $e = (a, b) \in E$, both endpoints may invest. The probability that an attacker survives the edge is:

$$S(e) = (1 - x_{a,e})(1 - x_{b,e}).$$

The interpretation is that each endpoint independently attempts interception, and the attacker survives the edge only if both attempts fail. This product structure captures the idea that defense on a shared connection compounds: even modest investments from both sides can produce substantial total protection.

A clarification is needed about when the formula is genuinely two-sided. If only one endpoint invests — because the other has no incentive to do so — then the formula collapses to the original Bloch survival expression. In particular, for the entrance edge $e_1 = (0, b_1)$ between the attacker and the first node, the attacker invests nothing, so $x_{0,e_1} = 0$ and $S(e_1) = (1 - x_{b_1,e_1})$, which is exactly Bloch's expression. Both endpoints contribute meaningfully only when both belong to the attacker's support and the edge lies on attack paths targeting each of them.

[Path survival probability] For a path $p = (e_1, e_2, \dots, e_L)$ from node 0 to a target $k(p)$, the probability that the attacker survives the entire path is the product of edge survival probabilities:

$$\alpha_{k(p)}(p) = \prod_{\ell=1}^{L} S(e_\ell) = \prod_{\ell=1}^{L} (1 - x_{a_\ell,e_\ell})(1 - x_{b_\ell,e_\ell}),$$

where $e_\ell = (a_\ell, b_\ell)$ is the ℓ-th edge on the path.

In Bloch's model, the path survival probability α and the success probability β are distinct: the attacker first survives the intermediate nodes (factor α) and then survives the target's own defense (factor $1 - x_{k(p)}$). In the edge-based model, the target's investment is already absorbed into the survival product, because the last edge on the path includes the target as one of its endpoints. There is no separate interception step at the target. Therefore $\alpha_{k(p)}(p) = \beta_{k(p)}(p)$ in this model.

[Defender cost] The total cost incurred by defender i is:

$$C_i = \sum_{e \in E(i)} \frac{x_{i,e}^2}{2}.$$

Each edge investment incurs its own quadratic cost. The cost is separable, which means that investing on one edge does not affect the marginal cost of investing on another. This separability matters for the analysis: it ensures that the first-order condition for each edge investment can be solved independently of the others.

The attacker's payoff is:

$$U(\pi, x) = \sum_{p \in \mathcal{P}} \pi(p) \cdot \alpha_{k(p)}(p) \cdot b_{k(p)},$$

and each defender's payoff is:

$$V_i(\pi, x) = -\sum_{\substack{p \in \mathcal{P} \\ k(p)=i}} \pi(p) \cdot \alpha_i(p) \cdot b_i \; - \sum_{e \in E(i)} \frac{x_{i,e}^2}{2}.$$

Both payoffs are written explicitly as functions of π (the attacker's mixed strategy) and x (the full profile of defender investments). It is worth noting that only π is mixed; defenders choose their investments deterministically.

For the cooperative version of the model, a single coordinator controls all edge investments and minimizes the total expected loss across the network:

$$\mathcal{L} = \sum_{i \in \Delta} \left[\prod_{\ell=1}^{L_i} (1 - y_{a_\ell, e_\ell})(1 - y_{b_\ell, e_\ell}) \right] r_i \cdot b_i + \sum_{e \in E} \sum_{j \in e} \frac{y_{j,e}^2}{2},$$

where Δ denotes the set of attacked targets, r_i the probability of attack on target i, and $y_{j,e}$ the coordinator's chosen investment by node j on edge e. The double sum in the cost term reflects that every edge is paid for from both sides.

2.4 Similarities with Bloch

Several elements of Bloch's analysis carry over to the modified model without modification, because they depend on properties of the attacker's choice rather than on where defense is placed. We collect them here briefly.

Existence

[Existence] The edge-based defense game admits a Nash equilibrium in mixed strategies.

The argument is the standard application of the Debreu–Fan–Glicksberg fixed-point theorem. Strategy spaces are compact and convex (probability distributions over a finite set of paths, and bounded boxes of investment vectors). Payoffs are continuous, since they are sums and products of continuous functions. Each player's payoff is quasi-concave in their own strategy: the attacker's payoff is linear in π, and each defender's payoff is concave in their own investments (linear in the loss term, concave in the cost term). All three conditions hold, so an equilibrium exists. The full argument is presented in the Appendix.

Pure Nash Equilibrium

[Pure NE conditions] A pure Nash equilibrium exists if and only if there is a target i such that

(i) $b_i(1-b_i) \geq b_j$ for all j such that there is a path from 0 to j along which node i is not on every route, and

(ii) $b_i \geq b_j$ for all j such that node i lies on every path from 0 to j.

In a pure equilibrium, the attacker concentrates entirely on a single target, and the defender of that target invests on the single edge through which the attack arrives. With one defender investing on one edge, the model collapses to the Bloch case for the relevant fragment, and the conditions are identical.

Lemmas Carrying Over

Three of the four key lemmas in Bloch's analysis depend only on the attacker's optimization and survive unchanged. Lemma 1 states that all targets in the attacker's support yield the same expected payoff U — otherwise the attacker would shift weight away from the worse ones. Lemma 3 states that when multiple support members could serve as predecessors for a given target, the attacker routes through the one with the lowest value, because that route offers the highest survival probability. Lemma 4 states that target values must strictly increase along any attack path in the support, because attacking a less valuable target later (after surviving more defense) is strictly worse than attacking a more valuable target earlier.

Uniqueness

[Uniqueness] Under the genericity assumption ($b_i \neq b_j$ for all $i \neq j$), the edge-based defense game admits a unique Nash equilibrium.

The argument follows Bloch's contradiction structure. If two equilibria existed with different supports, one support would be strictly contained in the other, the smaller support would correspond to a lower attacker payoff, and the attacker in the equilibrium with the smaller support would have a profitable deviation to a target outside that support. The contradiction collapses the assumption of multiple equilibria.

Deeper-Target Equilibrium Equations

For a deeper target i with immediate predecessor k in the support, the equilibrium investment and attack probability are:

$$x_{i,e_i} = 1 - \frac{b_k}{b_i}, \qquad q_i = \frac{b_k(b_i - b_k)}{U \cdot b_i^2}.$$

These are identical to Bloch's expressions, because a deeper target has only one relevant incoming attack edge — the connection to its lowest-value predecessor in the support. There is nothing to split, and the math collapses to the original case.

2.5 Differences from Bloch

The interesting changes happen elsewhere. We now describe each of them.

From One Variable to Many

In Bloch's model, defender i chooses a single investment x_i. In the edge-based model, defender i chooses a vector of investments, one per incident edge. The dimensionality of each defender's strategy space grows from 1 to $|E(i)|$, the degree of node i in the network. This change is mechanical, but it has consequences for everything that follows.

Two Interception Attempts Per Edge

Survival through a single edge is now $(1 - x_{a,e})(1 - x_{b,e})$ rather than $(1 - x_i)$. When both endpoints have an incentive to invest, both attempt interception independently, and the attacker has to escape both. This is the source of the new strategic interactions: the two ends of an edge now appear in each other's first-order conditions, and their choices are no longer independent.

Separable Cost Across Edges

In Bloch, defender i's cost is a single quadratic term $x_i^2/2$. In the edge-based model, it is a sum of independent quadratic terms across all incident edges. The convexity of each term is preserved, but the cost is now distributed across multiple decisions rather than concentrated in one. This separability is what makes the cost savings result possible in the cooperative analysis.

Modified Lemma 2

Bloch's Lemma 2 states that, without loss of generality, the attacker uses exactly one path to reach each target in the support. The proof rests on the fact that, in the node-based model, multiple paths to the same target collapse to equivalent survival probabilities — since the target's single investment covers all directions of approach, the attacker is indifferent between routes.

This argument breaks down in the edge-based model. If the attacker concentrated all attacks targeting node i on a single incoming edge, the defender would leave the other incoming edges undefended. The attacker would then deviate to one of the undefended edges, breaking the equilibrium. The only consistent outcome is for the attacker to mix across *all* free-highway incoming edges to a first target, and for the defender to invest equally on each of them.

This is the continuous analog of the mutual coverage condition in Mavronicolas et al. (2008). In their discrete world, the defender must protect every edge that could be attacked, and the attacker must threaten every edge that is defended; otherwise one side could improve. In the continuous world here, the same logic determines the equal-investment, equal-mixing structure on multiple incoming edges.

[Free-highway edge] An edge e incident to target i is a *free-highway edge* if there is a path from node 0 to i through e that passes only through nodes investing zero (nodes outside the support).

[Multi-path factor m_i] For a first target i, define

$$m_i = |\{e \in E(i) : e \text{ is a free-highway edge to } i\}|.$$

A clarification is essential here. The factor m_i counts *edges incident to the target*, not paths from the attacker. If multiple paths converge to a single edge before reaching target i, those paths share the same final edge and contribute $m_i = 1$, not more. The structural weakness comes from defending multiple distinct incoming edges, each requiring its own investment, not from the existence of multiple route options upstream.

First-Target Equilibrium Equations

For a first target i with m_i free-highway incoming edges:

$$x_{i,e} = 1 - \frac{U}{b_i}, \qquad q_i = \frac{m_i(b_i - U)}{b_i^2}.$$

The per-edge investment is the same as in Bloch's model. The attacker's indifference condition, applied to a single edge, demands the same payoff U regardless of how many other edges exist, so the per-edge defense level is unchanged. What changes is the total attack probability, which is multiplied by m_i. The defender now has to maintain this investment on m_i separate edges, and the attacker's total weight on the target rises proportionally.

It is worth noting that the feasibility condition $U/b_i < 1$ holds for all targets in the support: if $U \geq b_i$, then q_i would be non-positive and target i could not be in the support. The investment $x_{i,e} = 1 - U/b_i$ is therefore strictly positive at every supported target.

Multi-Path Targets Are Structurally Weaker

When $m_i > 1$, target i pays a total defense cost of

$$C_i = m_i \cdot \frac{(1 - U/b_i)^2}{2},$$

which is m_i times the cost in Bloch's model. The per-edge investment is unchanged, but the burden of maintaining it across multiple edges grows linearly with the number of incoming edges. Multi-path targets are therefore more expensive to defend, and the attacker exploits this by concentrating more attack probability on them. This is the core consequence of moving defense to edges.

Strategic Substitutes on Shared Edges

When both endpoints of an edge $e = (i, j)$ have incentive to invest — meaning the edge lies on attack paths targeting both nodes — their best responses interact directly. Defender i's best response on e is:

$$x^*_{i,e} = q \cdot b_i \cdot (1 - x_{j,e}),$$

where q is the attack probability flowing through e targeting i. Differentiating with respect to $x_{j,e}$ gives:

$$\frac{\partial x^*_{i,e}}{\partial x_{j,e}} = -q \cdot b_i < 0.$$

This is strategic substitutes: when one endpoint invests more, the other's optimal response is to invest less. The mechanism is intuitive — if my neighbor is already catching most attackers on the connection we share, my own marginal investment matters less — and it is entirely absent from Bloch's model, where each node's investment never enters another node's decision.

A natural concern is whether this dynamic could push both endpoints to invest nothing, leaving the shared edge undefended. It cannot. If one endpoint invests zero, the other's best response is $q \cdot b_i$, which is strictly positive whenever attacks flow through the edge. The pair $(0, 0)$ cannot be part of any equilibrium. The real questions about strategic substitutes — whether the interior equilibrium is unique, whether corner solutions arise, whether the equilibrium is stable — are open and are discussed at the end of this report.

To illustrate why this mechanism rarely shows up on simple networks, consider a line ATK $\to T_1 \to T_2 \to T_3$. The edge (T_1, T_2) lies on the attack path to T_2 and to T_3, but not to T_1, because the attacker has already passed T_1 by the time it traverses this edge. Defender T_1 has no incentive to invest on (T_1, T_2), since attacks through this edge do not target T_1. The same is true for every interior edge: only the downstream node has reason to invest. Strategic substitutes therefore do not arise on lines, and they appear only in networks where a single edge serves as the attack route for multiple downstream targets.

Braess Paradox Does Not Occur

In the original model, adding a new connection to the network can sometimes harm the attacker, because the additional vulnerability prompts defenders to increase their investments more than enough to offset the new attack route. This is a version of Braess's paradox.

In the edge-based model, this paradox does not occur. Adding a new free-highway edge to a first target i raises m_i by one. Through implicit differentiation of the equilibrium constraint $\sum_{j\in\Delta} q_j(U) = 1$, one obtains:

$$\frac{dU}{dm_i} = \frac{(b_i - U)/b_i^2}{\sum_{j\in\Delta} |\partial q_j/\partial U|} > 0.$$

The derivative is strictly positive: adding a connection always benefits the attacker. The reason is that each new edge must be defended separately at the same per-edge level, but the attacker now has more attack vectors, and the per-edge defense cannot rise to compensate because it is already pinned down by the attacker's indifference condition. Braess's paradox in its classical form simply has no room to operate.

Cooperative Symmetry and Cost Savings

Two distinctive results emerge in the cooperative version of the model. The first is symmetry within edges.

[Symmetry within edges] In any cooperative equilibrium, both endpoints of an edge $e = (i, j)$ invest equally: $y_{i,e} = y_{j,e}$.

The argument is short. The two investments $y_{i,e}$ and $y_{j,e}$ enter the cooperative loss function in exactly the same way — both multiply $(1-y)$ inside the survival product, and both have identical quadratic cost terms. The first-order conditions are therefore symmetric in the two variables, and the unique solution sets them equal. The full derivation, using either subtraction of FOCs or substitution, is in the Appendix.

The second result is cost savings on lines.

[Cost savings under shared support] On a line network, when the cooperative equilibria under node-based and edge-based defense share the same support, the two formulations achieve the same attacker utility U, but the centralized defender incurs strictly lower total investment cost under edge-based defense.

The intuition is the convexity of the cost function. Splitting a total investment of y between two endpoints, each contributing $y/2$, gives a total cost of $2 \cdot (y/2)^2/2 = y^2/4$, which is strictly less than the single-node cost of $y^2/2$. The same level of effective interception can be achieved with two smaller investments at lower combined cost than with one larger investment, and the cooperative defender exploits this systematically across every position on the line.

2.6 Numerical Illustrations

To make the effects concrete, we work through two numerical examples. The first illustrates the cost savings that arise on a line network when the support is the same under both defense models. The second illustrates multi-path weakness on a small general network.

Example 1: A Line of Three Targets (Cooperative Comparison)

Consider the network ATK $\to T_1 \to T_2 \to T_3$ with values $b_1 = 0.40$, $b_2 = 0.50$, $b_3 = 0.70$. Under cooperative defense, the centralized defender finds it optimal to drop T_2 from the attack support and use it as a shield. The result is the same attacker utility under both defense models, $U = 0.24$, but with different investment structures shown in Table 2.1.

	Node Defense (Bloch)	**Edge-Based**
Attacker utility U	0.2400	0.2400
Support	$\{T_1, T_3\}$	$\{T_1, T_3\}$
r_1 (attack probability on T_1)	0.2313	0.5264
r_3 (attack probability on T_3)	0.7687	0.4736
Investment at position 1	0.4000	0.4000 (entrance edge)
Investment at position 2	0.2441 (one node)	0.1307 (per endpoint)
Investment at position 3	0.2441 (one node)	0.1307 (per endpoint)
Effective interception, position 2	0.2441	$1 - (1 - 0.1307)^2 = 0.2441$
Effective interception, position 3	0.2441	0.2441
Total defense cost	0.1396	0.1142
Cost saving	—	18.2%

Table 2.1: Cooperative comparison on a line network with values $(0.40, 0.50, 0.70)$.

The same level of effective interception is achieved at every position. What changes is how that interception is paid for. Under node defense, a single defender bears the full cost of $0.2441^2/2 = 0.0298$ at position 2, and the same again at position 3. Under edge defense, each interior edge receives investment from two endpoints, each paying $0.1307^2/2 = 0.00854$, so the cost per interior edge is 0.0171 rather than 0.0298. The convexity of the quadratic cost function means that splitting investment across two payers is genuinely cheaper than concentrating it on one. The total saving aggregates to 18.2% across the line.

The attack probabilities differ between the two models because the cooperative defender's first-order conditions on interior edges have a different structure under edge defense (involving an additional $(1-x)$ factor from the survival product), but the equilibrium attacker utility is the same in both cases in this example, since the support coincides under both defense models. This is the cleanest demonstration of the cost-savings result: the attacker is no better or worse off, but defense is delivered more efficiently.

Example 2: A General Network with a Multi-Path Target (Decentralized Comparison)

Consider a small general network: the attacker ATK connects to T_1 and T_2; T_1 connects to T_3 and T_4; T_2 connects to T_4. Values are $b_1 = 0.20$, $b_2 = 0.25$, $b_3 = 0.50$, $b_4 = 0.70$. Target T_4 has two distinct incoming free-highway edges — one from T_1 and one from T_2 — so $m_4 = 2$. Target T_3 has only one incoming edge, so $m_3 = 1$. Targets T_1 and T_2 are first targets reachable directly from the attacker.

	Node Defense (Bloch)	**Edge-Based**
Attacker utility U	0.4020	0.4773
Change in U	—	+18.7%
Support	$\{T_3, T_4\}$	$\{T_3, T_4\}$
m_3	1	1
m_4	1	2
Attack probability q_3	0.3920	0.0909
Attack probability q_4	0.6081	0.9091
Investment x_3	0.1960	0.0455
Investment x_4	0.4257	0.3181 (per edge)
T_3 defense cost	0.0192	0.0010
T_4 defense cost	0.0906	0.1012

Table 2.2: Decentralized comparison on a general network with values $(0.20, 0.25, 0.50, 0.70)$.

The story behind these numbers is what matters more than the numbers themselves.

The mechanism begins with T_4. Under node defense, T_4 has one investment that covers all directions of approach, and the attacker faces full interception strength regardless of which path is used. Under edge defense, the same target must defend two separate incoming edges, each receiving its own investment. The per-edge investment formula gives $x_{4,e} = 1 - U/b_4$ on each edge, but the total attack probability now scales with $m_4 = 2$.

This raises the attacker's expected payoff U, because T_4 is structurally weaker. The rise in U propagates through the rest of the network. With U now larger, the equilibrium investments at other targets adjust: T_3 in particular finds that defending itself becomes barely worthwhile, since its value $b_3 = 0.50$ is only slightly above the new $U = 0.4773$. The investment at T_3 collapses from 0.1960 to 0.0455.

The attacker, facing a much weaker T_4 with two routes of approach, concentrates almost all attacks there: q_4 rises from 61% to 91%. T_3 becomes a secondary target. The total cost picture is more nuanced — T_4's cost rises slightly because it is now defending two edges, while T_3's cost collapses because it is barely defending at all.

The logical chain is therefore: more incoming edges to a target $\rightarrow$ each edge structurally weaker $\rightarrow$ higher U $\rightarrow$ lower-value targets pushed toward the margin $\rightarrow$ attacks concentrate on the multi-path target. Each step follows from the previous one, and the entire chain is a direct consequence of moving defense to edges. The simulator produces the same numbers when the network is built and solved under each defense mode, providing a numerical check on the analytical results.

2.7 Two Further Questions

During the supervision, two additional questions arose. We address them briefly here, both because they sharpen the picture of the model and because they illustrate the kind of variations a future researcher might want to study.

Joint Cost on Shared Edges

The model assumes that each endpoint of a shared edge pays its own cost independently: $C_i = \sum_e x_{i,e}^2/2$. An alternative would be a joint cost, in which the total investment on an edge is what enters the cost function: $C(e) = (x_{a,e} + x_{b,e})^2/2$. Under this alternative, defender i's best response on a shared edge becomes:

$$x_{i,e}^* = q \cdot b_i \cdot (1 - x_{j,e}) - x_{j,e}.$$

Compared to the separable-cost expression, an additional $-x_{j,e}$ term appears at the end. The neighbor's investment now reduces the optimal response in two ways: through the survival term (if my neighbor catches more, my marginal benefit shrinks) and through the cost term (if my neighbor invests more, my marginal cost rises). The free-riding pressure is intensified.

The implication is that joint cost would amplify the strategic substitutes effect, leading to even lower total defense on shared edges and a stronger position for the attacker. The simple structure of the modified best response makes this a tractable variation that could be developed into a fuller analysis without changing the model's overall framework.

Imperfect Information

A second question is what happens if defenders only know about their immediate neighbors — the nodes they share an edge with — rather than the entire network. This restriction sounds substantial but turns out not to change the analysis. The only place where one defender's choice directly enters another's first-order condition is on shared edges, where strategic substitutes operate. By assumption, neighbors share an edge and therefore observe each other under the imperfect information condition. Defenders that do not share an edge do not interact directly; their connection runs only through the attacker's mixed strategy, which is determined by the attacker's indifference conditions and does not require any defender to observe distant nodes.

In short, the information restriction does not bind on the strategic interactions that matter. The equilibrium is unchanged. This is a useful observation in its own right, because it shows that the model's predictions are robust to limitations on what individual defenders can observe.

Conclusion to the Chapter

This chapter has built the modified model from the ground up, established its formal properties, and illustrated its behavior on small examples. The structural changes are concentrated around three places: the survival formula on individual edges, the new factor m_i that captures multi-path weakness at first targets, and the strategic substitutes mechanism on shared edges. From these three changes, the rest follows: amplified responses to value changes, the disappearance of Braess's paradox, the symmetry of cooperative defense within edges, and the cost savings on lines. Two further questions — joint cost and imperfect information — show how the framework can be extended or restricted without losing tractability. The implementation that supports these results numerically is the subject of the next chapter.

Chapter 3

Implementation and Complementary Contributions

Introduction to the Chapter

Chapter 2 established the theoretical results of the modified model. This chapter is about everything else: the working tool that was built to verify those results numerically, the skills that the project developed along the way, and the experience of doing research independently for the first time. The chapter is shorter than the previous one, and intentionally so. Theory is the centerpiece of the work; implementation and reflection are what made the theory possible to produce and check.

3.1 From Theory to Working Code

The motivation for building a simulator came early in the project, and it was practical rather than ambitious. Working through the equilibrium of a four-target network by hand takes about an hour. Working through a seven-target network with multiple paths takes the better part of a day, and the answer is then almost impossible to verify without redoing the entire computation from scratch. By the time the model needed to be tested on networks large enough to demonstrate the multi-path effect clearly, hand calculation had become a bottleneck.

A simulator removes that bottleneck. It also does something else, which turned out to be more important. It makes the theory testable in a way that printed equations are not. Once the model is in code, every claim made in Chapter 2 can be checked against numerical output. The condition that target values strictly increase along an attack path, the equilibrium equations with the multi-path factor m_i, the cost-savings result on lines, the disappearance of Braess's paradox — each of these was verified by running the simulator on representative networks and confirming that the numerical output matched the analytical prediction. When something failed to match, the disagreement always pointed to a real error somewhere, in the code or in the math or in my own understanding. This kind of feedback loop is hard to produce any other way.

The choice of R Shiny as the platform was deliberate. R has strong libraries for graph manipulation (`igraph`) and visualization (`ggraph`), which made the network display layer straightforward to build. Shiny adds a web interface on top, so the simulator can be deployed publicly and accessed from any browser without installation. The complete tool is hosted at https://chukwudi-ogbuta.shinyapps.io/blochnetwork/ and runs without any setup on the user's side.

3.2 How the Simulator Works

The application is organized into three tabs that follow the natural flow of investigation: build a network, solve its equilibrium, and run experiments on it.

The Network Builder

The first tab lets the user construct a network from scratch. The user chooses the network type (line, star, or general), specifies the number of targets, and enters a value for each target. For general networks, an edge selector lets the user click on which pairs of nodes are connected. The result is rendered as a graph, with nodes colored by value and the attacker shown in red.

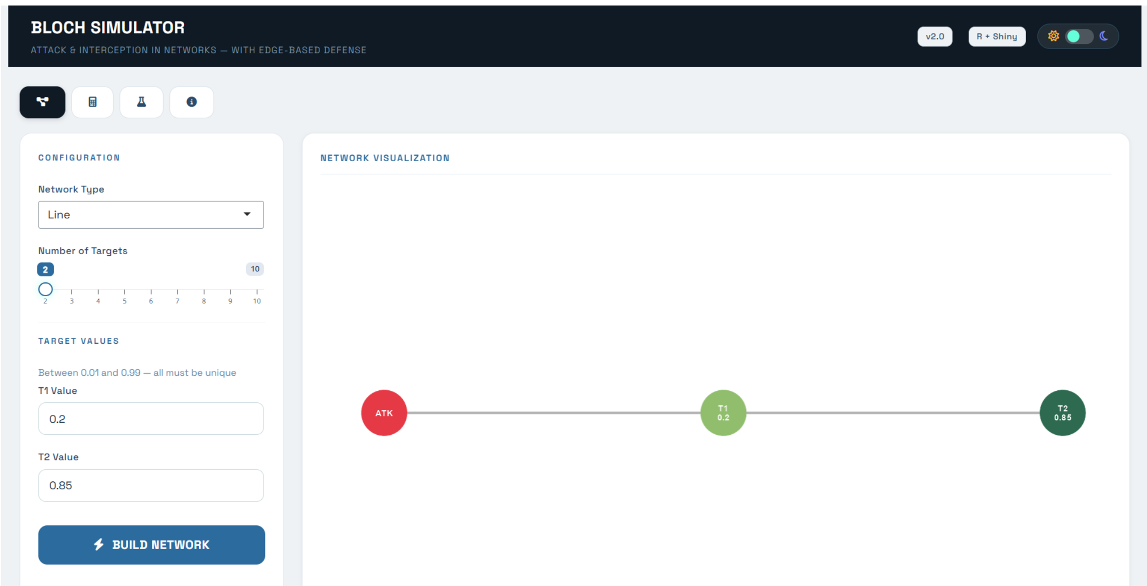


Figure 3.1: The Network Builder tab. The user specifies the network structure and target values; the simulator displays the resulting graph.

The builder enforces basic consistency checks before allowing the network to be used: target values must be unique (the genericity assumption), every target must be reachable from the attacker, and no isolated nodes are allowed. These checks catch mistakes before they propagate into the solver.

The Equilibrium Solver

The second tab is where the analytical results meet the keyboard. The user selects a defense model (node-based as in Bloch's original formulation, or edge-based as in the modification), chooses between decentralized and cooperative defense, and runs the solver. The simulator returns the attacker's expected utility U, the equilibrium support, the attack probabilities q_i, and the per-edge investments. A side-by-side comparison view runs both defense models on the same network and presents the results next to each other.

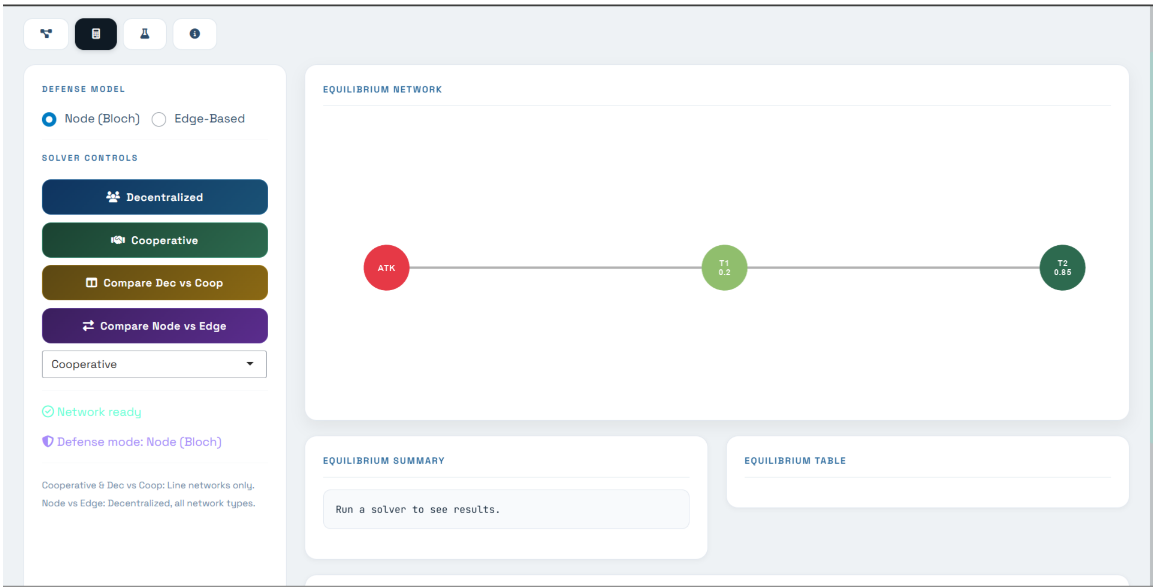


Figure 3.2: The Equilibrium Solver tab. The user picks a defense model and solver type; the simulator returns the equilibrium values, attack probabilities, and per-edge investments.

The solver implements the iterative support-reduction procedure from Chapter 2. It starts by assuming all targets are in the support, computes the equilibrium under this assumption, and removes any target whose attack probability comes out non-positive. It repeats until every remaining target has a strictly positive attack probability. For the edge-based model, the solver computes the multi-path factor m_i for each surviving target by counting the free-highway incoming edges, and applies the modified equilibrium equations.

The Experimentation Tab

The third tab implements the comparative statics from Chapter 2 directly. The user picks an experiment from a menu — changing a target's value, adding a new connection, or removing a target from the attacker's consideration — specifies the parameters, and runs it. The simulator solves the equilibrium twice, once on the original network and once on the modified version, and presents the results side by side along with an impact summary that quantifies the change in attacker utility.

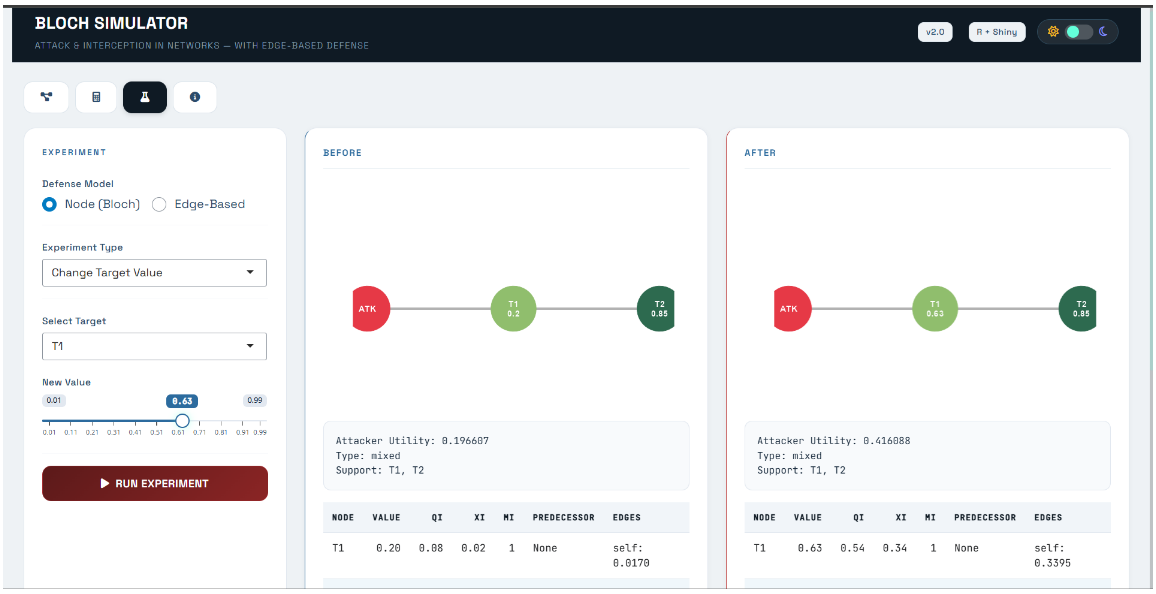


Figure 3.3: The Experimentation tab. The user perturbs the network and sees how the equilibrium shifts.

This tab is the part of the simulator most directly connected to the report's analytical chapter. It turns the comparative statics into something the user can run rather than just read about.

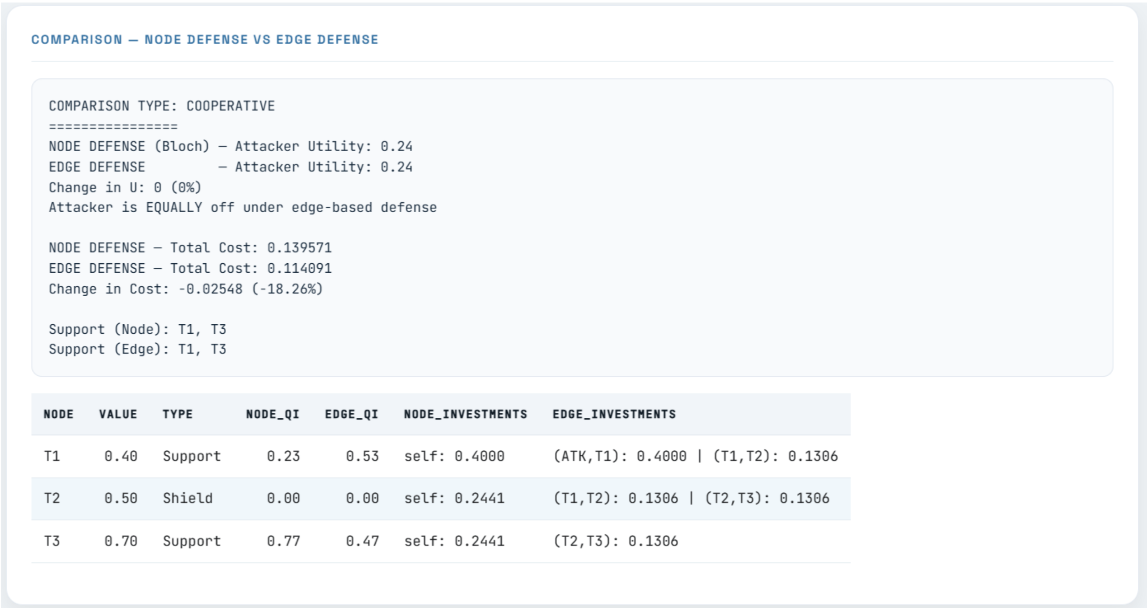
3.3 Numerical Verification of the Examples in Chapter 2

The two worked examples in Chapter 2 served as the primary test cases for the simulator. Their numerical predictions can be confirmed directly in the application.

Example 1: Cooperative Comparison on a Line

The line network with values $(0.40, 0.50, 0.70)$ was constructed in the Network Builder and solved using the cooperative comparison feature. The output reproduced the values stated in Chapter 2: attacker utility $U = 0.24$ in both defense models, with total cost falling from 0.1396 under node defense to 0.1142 under edge defense, a saving of 18.2%. The equilibrium table displays the per-edge investments under edge mode, making explicit that node T_2 invests on each of its two adjacent edges and that the total cost on each interior edge reflects two endpoints contributing equally.

Figure 3.4: Example 1: a line network of three targets with values $(0.40, 0.50, 0.70)$.



COMPARISON — NODE DEFENSE VS EDGE DEFENSE

```
COMPARISON TYPE: COOPERATIVE
================
NODE DEFENSE (Bloch) — Attacker Utility: 0.24
EDGE DEFENSE         — Attacker Utility: 0.24
Change in U: 0 (0%)
Attacker is EQUALLY off under edge-based defense

NODE DEFENSE — Total Cost: 0.139571
EDGE DEFENSE — Total Cost: 0.114091
Change in Cost: -0.02548 (-18.26%)

Support (Node): T1, T3
Support (Edge): T1, T3
```

NODE	VALUE	TYPE	NODE_QI	EDGE_QI	NODE_INVESTMENTS	EDGE_INVESTMENTS
T1	0.40	Support	0.23	0.53	self: 0.4000	(ATK,T1): 0.4000 | (T1,T2): 0.1306
T2	0.50	Shield	0.00	0.00	self: 0.2441	(T1,T2): 0.1306 | (T2,T3): 0.1306
T3	0.70	Support	0.77	0.47	self: 0.2441	(T2,T3): 0.1306

Figure 3.5: Example 1 solution: side-by-side cooperative comparison under node and edge defense.

The match between the analytical prediction and the simulator output confirms the cost-savings result derived in Chapter 2. Same level of effective interception at every position, lower total cost under edge defense, and the difference traceable directly to the convexity of the quadratic cost function.

Example 2: Decentralized Comparison on a General Network

The four-target general network with values $(0.20, 0.25, 0.50, 0.70)$ tests the multi-path weakness result. Target T_4 has two incoming edges, one through T_1 and one through T_2, both free-highway since T_1 and T_2 are not in the support. The simulator confirms the predicted equilibrium: under node defense, $U = 0.4020$; under edge defense, $U = 0.4773$. The change in U of $+18.7\%$ reflects the structural weakness introduced by the multi-path factor $m_4 = 2$. Attack probability on T_4 rises from 61% to 91%, and the investment at

T_3 collapses from 0.196 to 0.045 as T_3 becomes nearly indifferent to attack at the higher equilibrium U.

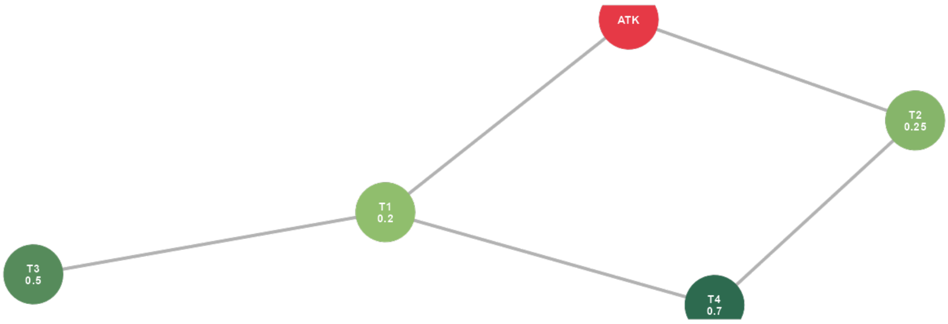
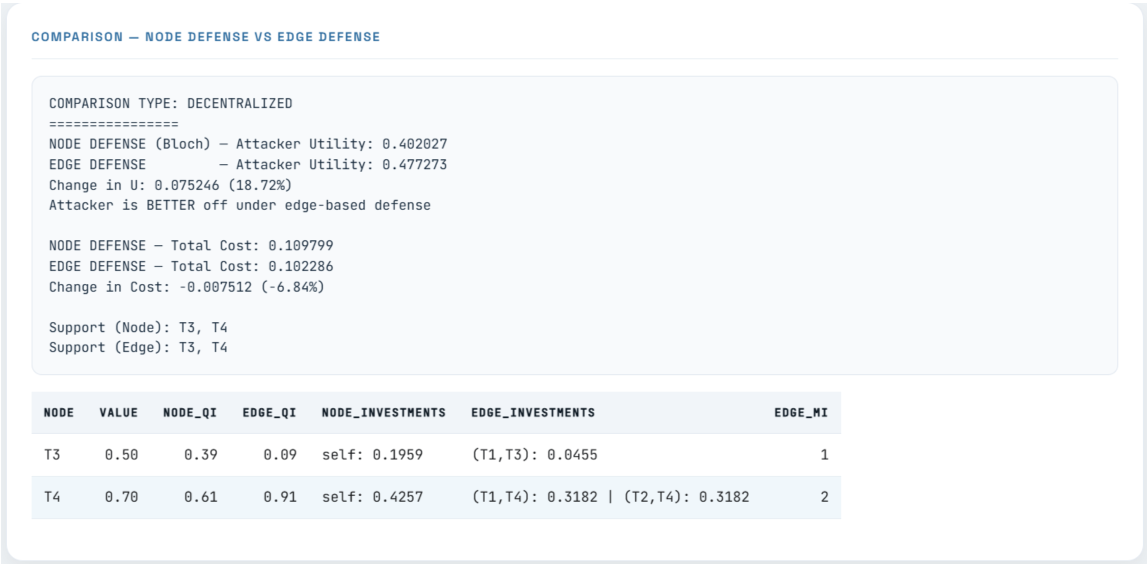


Figure 3.6: Example 2: a general network with T_4 reachable through two distinct paths.



COMPARISON — NODE DEFENSE VS EDGE DEFENSE

```
COMPARISON TYPE: DECENTRALIZED
================
NODE DEFENSE (Bloch) — Attacker Utility: 0.402027
EDGE DEFENSE         — Attacker Utility: 0.477273
Change in U: 0.075246 (18.72%)
Attacker is BETTER off under edge-based defense

NODE DEFENSE — Total Cost: 0.109799
EDGE DEFENSE — Total Cost: 0.102286
Change in Cost: -0.007512 (-6.84%)

Support (Node): T3, T4
Support (Edge): T3, T4
```

NODE	VALUE	NODE_QI	EDGE_QI	NODE_INVESTMENTS	EDGE_INVESTMENTS	EDGE_MI
T3	0.50	0.39	0.09	self: 0.1959	(T1,T3): 0.0455	1
T4	0.70	0.61	0.91	self: 0.4257	(T1,T4): 0.3182 | (T2,T4): 0.3182	2

Figure 3.7: Example 2 solution: side-by-side decentralized comparison under node and edge defense.

The simulator output reproduces every quantity reported in Chapter 2's Table 2.2, providing an independent numerical check on the analytical derivation.

3.4 A Worked Comparative Experiment

To illustrate how the experimentation tab connects the theoretical analysis to interactive use, consider the following comparative-statics experiment on the general network from Example 2. The original network has T_4 reachable through two distinct free-highway edges, giving multi-path factor $m_4 = 2$. What happens if a new connection is added between T_3 and T_4?

Adding this edge changes the structure of attacks on T_4. Previously T_4 had two incoming free-highway edges; now it has three, since T_3 is also not in the support of the modified equilibrium and the path ATK $\to T_1 \to T_3 \to T_4$ becomes a third free-highway route. The multi-path factor m_4 rises from 2 to 3.

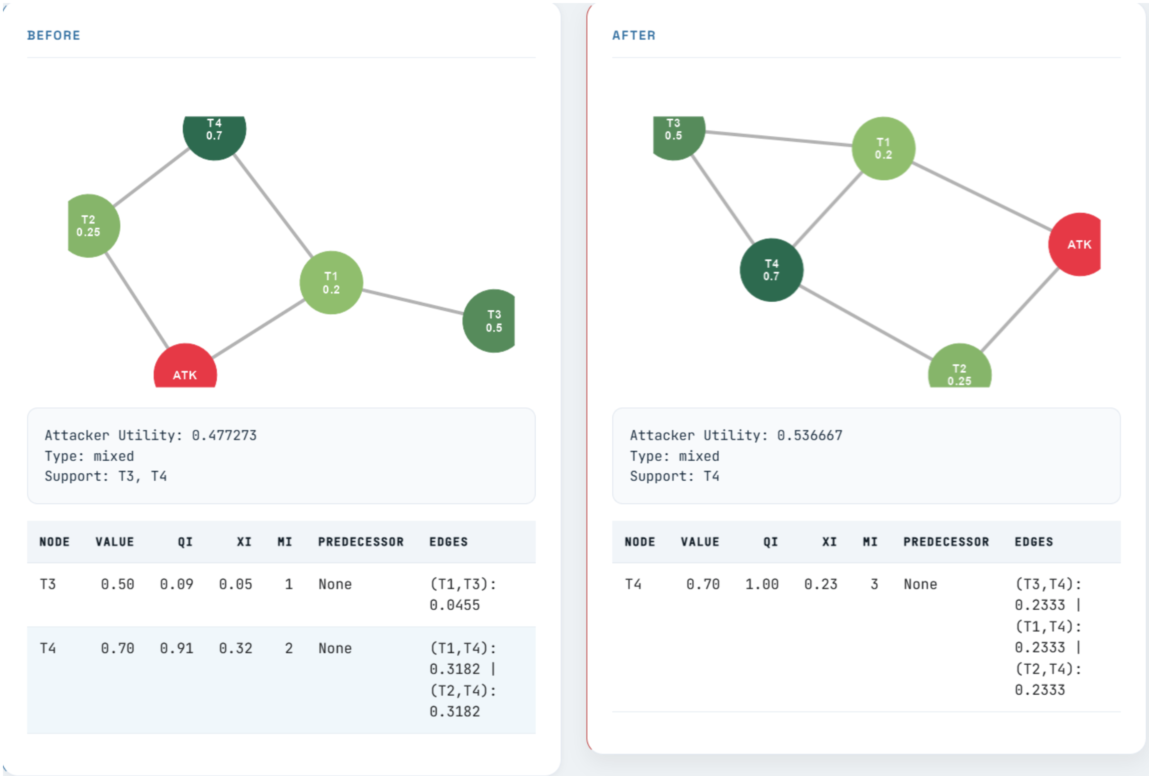


Figure 3.8: The comparative experiment: adding edge (T_3, T_4) to the general network. The simulator solves the equilibrium before and after the modification.

The simulator output makes the consequences explicit. The attacker's expected utility rises from 0.4773 to 0.5367, an increase of about 12.4%. Target T_3, which had a small but positive attack probability before the modification, drops out of the support entirely after the new edge is added. All attack probability concentrates on T_4, which now defends three incoming edges at a per-edge investment of 0.2333 each. The total cost picture shifts as well: T_3 no longer pays anything, while T_4's burden grows because it must defend an additional edge.

The reasoning behind the result is the same logical chain identified in Chapter 2. More incoming free-highway edges to a target means each edge is structurally weaker, since the per-edge investment formula $x = 1 - U/b_i$ is unchanged but must now be maintained on more edges. The aggregate effect raises U, which pushes lower-value targets out of the support, which in turn concentrates the attacker's probability on the multi-path target. The experiment thus confirms the theoretical claim that adding a connection to a multi-path target unambiguously benefits the attacker — a result that distinguishes the edge-based model from Bloch's, where Braess-type paradoxes are possible.

This experiment is the kind of investigation that would have taken hours to do by hand and now takes about ten seconds in the simulator. The interactive format also makes it easier to explain the result to others. Showing a supervisor the before-and-after view conveys the dynamic in a way that a static table of numbers does not.

3.5 Skills and Working Methods

Beyond the theoretical and computational outputs, the internship developed a set of skills that I want to record honestly rather than as a list of bullet points.

Technical Skills

The most concrete additions to my technical toolkit were the formal mathematics of game theory and the corresponding software workflow. Before this internship I had taken game theory courses at the master's level, but I had never derived a Nash equilibrium from first principles, never written out a proof of existence using the Debreu-Fan-Glicksberg framework, and never worked through comparative statics by implicit differentiation on an equilibrium equation. Doing each of these for the first time, with corrections from a supervisor who has spent his career in cooperative game theory, taught me how the formal apparatus actually works.

The software side developed similarly. I had built Shiny applications before, but never one driven by network structures and graph computations. The integration of `igraph` for network logic with `ggraph` for visualization required a pattern of separating computation from display that I had not worked with before. Writing the simulator also forced me to internalize the structure of the equilibrium algorithm. There is a difference between understanding a procedure on paper and being able to write code that executes it correctly; the simulator was the place where that difference became visible to me.

LaTeX was familiar from earlier coursework, but this project pushed it further. The proofs document, the presentation slides, and the report itself all required structures more elaborate than anything I had written before — proper theorem environments, bibliographies, cross-references, tables with consistent formatting. The discipline of producing professional academic documents in LaTeX is now a habit rather than an effort.

Analytical Skills

The most important analytical lesson was about how research questions are actually structured. Going in, I imagined the work would proceed in a straight line: take the model, change one feature, derive the consequences. In practice the work involved a constant tension between making changes that were small enough to remain tractable and making changes that were substantive enough to be interesting. Several variations were considered and abandoned because their consequences became impossible to characterize cleanly. The variation that became the focus of the report was selected as much for its analytical clarity as for its conceptual interest.

Working through this taught me to distinguish between a question that is hard because it is genuinely deep and a question that is hard because it is poorly posed. A good model adjustment is one where each change has a traceable consequence, and where the resulting analysis can be carried out without resorting to numerical methods or special-case arguments. The edge-based modification fits this description; several alternatives I tried did not.

Working Methods and Cross-Validation

A practical habit that the internship reinforced was the use of multiple tools to cross-check results. Throughout the project I worked with Claude (for writing, derivation, and code), DeepSeek (for paper analysis and second opinions on derivations), and the simulator I built (for numerical verification). When all three agreed on a result, I had high confidence in it. When they disagreed, the disagreement itself was informative — one of them was wrong, and the discipline of figuring out which one taught me more than getting the right answer the first time would have. There were several occasions during the project where one of the tools produced an answer that turned out to be incorrect; the cross-validation caught these before they made it into the report.

This habit generalizes. Trusting any single source — including one's own first calculation — is fragile. Building the practice of independent verification, whether through a second tool, a different method of derivation, or a numerical check, is one of the most useful working methods I take from this internship.

3.6 A Reflection on the Process

The largest difference between this internship and the work I had done before was the absence of a fixed structure. A coursework assignment comes with a question, an expected method, and a known approximate length of time to completion. A research internship is different. The question itself was open-ended at the start, and a substantial part of the early work was understanding the existing literature well enough to discuss directions productively with my supervisor. The edge-based modification was suggested by Professor Béal as a promising angle, and once we discussed how it would change the model's structure, the value of pursuing it was clear. The early weeks were less about finding the right question and more about acquiring the technical vocabulary to engage with it seriously.

Working independently for extended periods also developed a discipline I had not built before. Without a deadline at the end of every week, the temptation to delay difficult parts is strong. The countermeasure I found most useful was setting myself small, specific milestones — a single proof to write, a single example to compute, a single section of code to refactor — and treating each of them as a unit of work that could be completed in a sitting. The cumulative effect of this approach was that the report grew gradually rather than in panicked bursts.

A related lesson was about how to use a supervisor effectively. Early in the internship I tended to bring my supervisor finished work for review. This was inefficient: by the time the work was finished, the underlying choices had already been made, and any feedback came too late to change them. Later in the internship I learned to bring questions and partial work earlier, when the direction was still mutable. The conversations about the model setup, the edge-based modification, the role of m_i, and the joint-cost variation were all most productive when I came in with a half-formed idea rather than a polished one. This is a habit I expect to carry forward.

Conclusion to the Chapter

The simulator described in this chapter is not a side project to the theoretical work but a continuous part of it. Every result in Chapter 2 was checked numerically, and several errors in early drafts of the analysis were caught by the simulator before they made it into writing. The skills developed during the internship — in formal mathematics, in software, in academic writing, in working independently — are the practical companion to the theoretical contribution. Together they constitute what the internship produced.

General Conclusion

Summary of the Work

This report set out to answer a single question: when defense in a network attack game is moved from the locations to the connections between them, how does the strategic situation change? The starting point was the model of Bloch, Chatterjee and Dutta (2023), and the inspiration for the modification came from Mavronicolas et al. (2008), who showed in a discrete setting that placing defense on edges produces equilibria with a recognizably different structure.

The work proceeded in three connected parts. The first part, presented in Chapter 1, situated the contribution within two parallel traditions in the literature on network security games: the node-defense tradition that Bloch's model belongs to, and the smaller edge-defense tradition that has so far worked only with discrete, centralized formulations. The gap between them — a continuous, decentralized model with defense on edges — is what this report attempts to fill.

The second part, presented in Chapter 2, built the modified model from the ground up and derived its formal results. The model retains Bloch's continuous-investment framework but places each defender's investment on the edges incident to its node rather than at the node itself. Each edge is thus jointly defended by its two endpoints, with survival through the edge given by the product of the two endpoints' interception probabilities. From this single change, several consequences follow. Some elements of Bloch's analysis carry over: the equilibrium exists, is unique under the standard genericity assumption, and the equations governing deeper targets remain the same. Other elements change in ways that are structurally informative. Targets reachable through multiple incoming free-highway edges are penalized by a factor m_i that captures the number of such edges; this multi-path weakness raises the attacker's expected utility and concentrates attack probability on multi-path targets. Edges shared by two support members exhibit strategic substitutes, with each defender's marginal incentive to invest declining in the neighbor's investment. The cooperative version of the model produces two distinctive results: investments are symmetric within each edge, and on line networks the same level of attacker harm is achieved at strictly lower defense cost than under Bloch's node-based cooperative.

The third part, presented in Chapter 3, made the theory testable. An interactive R

Shiny simulator was built to verify the analytical results numerically, support comparative experimentation, and produce the comparison tables that appear in Chapter 2. The simulator confirms each numerical claim made in the analytical chapter and allowed several errors in early drafts of the analysis to be caught before they made it into the final report.

Limitations

Several limitations of the work deserve acknowledgment. The cooperative analysis is fully developed only for line networks; the extension to general networks was attempted but not fully characterized within the time available, and the difficulty of doing so analytically suggests that a tractable closed-form characterization may not be possible without further structural assumptions. The strategic substitutes mechanism on shared edges raises questions about uniqueness and stability of the interior equilibrium that this report does not resolve — the existence and characterization of equilibria when both endpoints of an edge belong to the support is established, but a full game-theoretic analysis of how the two endpoints arrive at a stable interior solution remains open. The joint-cost variation discussed at the end of Chapter 2 was sketched rather than fully analyzed; its detailed treatment is a natural extension that would require its own dedicated study.

The report also restricts attention to a single attacker. Multi-attacker extensions, in which several adversaries operate simultaneously on the same network, would substantially complicate both the equilibrium structure and the cooperative analysis, and were beyond the scope of this work.

Directions for Future Research

Several directions for future work follow naturally from the analysis presented here. A characterization of cooperative defense on general networks — moving beyond lines — would extend the cost-savings result to settings closer to real infrastructure. The asymmetric joint-cost variation, in which the two endpoints of a shared edge contribute under a single combined cost function rather than separable individual costs, presents a clean variation that the framework can handle and would amplify the strategic substitutes effect. Multi-attacker extensions, network design questions (where does the defender place new connections to minimize attacker payoff?), and stability analysis of the strategic-substitutes interior equilibrium each represent substantive research programs that build on the modification developed here.

A practical direction, complementary to the theoretical extensions, is the application of the model to specific infrastructure systems. The motivating examples in this report — power grids, water distribution networks, cross-border environmental infrastructure — have data structures rich enough to support empirical calibration of the model's parameters. A full case study on any one of these systems would test whether the qualitative predictions of the model survive contact with real-world data and whether the cost-savings result on lines extends meaningfully to more complex network shapes.

Personal Reflection

For me as a student, the most valuable outcome of this internship was not any single technical result. It was the experience of working on a research problem with no fixed structure, no predetermined method, and no guaranteed answer. Coursework had prepared me for exercises with known solutions; this internship taught me what it feels like to work on something where the question itself was open at the start, where several promising directions had to be abandoned along the way, and where the right answer had to be earned rather than given.

The discipline of cross-validating results across multiple tools, the habit of bringing partial work to a supervisor early rather than finished work too late, the practice of building a numerical check for every analytical claim — these are working methods I expect to carry forward into whatever I do next. The internship also clarified for me that the field of economic intelligence, which sits at the intersection of strategic analysis, formal modeling, and quantitative tools, is a direction I want to continue pursuing.

I am grateful to Professor Sylvain Béal for the opportunity to undertake this work and for the supervision that shaped it, and to CRESE and the BDEEM program for the institutional and intellectual environment that made it possible.

Appendix A

Formal Proofs

This appendix collects the formal proofs of the results stated in Chapter 2. Each proof is preceded by a short explanation of the strategy used, so that a reader interested in the reasoning rather than the algebraic details can follow the structure without working through every line. The notation is identical to that established in Chapter 2: $G = (V, E)$ is the network, $V = \{0, 1, \dots, n\}$ with 0 the attacker, $b_i \in (0, 1)$ the value of target i, π the attacker's mixed strategy over paths, q_i the induced attack probability on target i, $x_{i,e}$ the investment by node i on edge e, and $E(i)$ the set of edges incident to node i.

A.1 Existence of Equilibrium

[Existence] The edge-based defense game admits a Nash equilibrium in mixed strategies.
Strategy. The proof applies the Debreu-Fan-Glicksberg fixed-point theorem. We show that strategy spaces are compact and convex, that payoffs are continuous, and that each player's payoff is quasi-concave in their own strategy. These three conditions together guarantee the existence of a Nash equilibrium.

Proof. The attacker's strategy space is the set of probability distributions over the finite set of paths $\mathcal{P}$, which is the standard simplex of dimension $|\mathcal{P}|$. This set is non-empty, compact, and convex.

For each defender i, the strategy space is $[0, 1]^{|E(i)|}$, the unit hypercube of dimension equal to the number of edges incident to i. This set is also non-empty, compact, and convex.

The attacker's payoff function is

$$U(\pi, x) = \sum_{p \in \mathcal{P}} \pi(p) \cdot \alpha_{k(p)}(p) \cdot b_{k(p)},$$

where $\alpha_{k(p)}(p) = \prod_{e \in p}(1 - x_{a,e})(1 - x_{b,e})$ for $e = (a, b)$. This is a polynomial in π and x, hence continuous.

The attacker's payoff is linear in π for fixed x, so it is automatically quasi-concave in the attacker's own strategy.

Each defender's payoff is

$$V_i(\pi, x) = -\sum_{\substack{p \in \mathcal{P} \\ k(p)=i}} \pi(p) \cdot \alpha_i(p) \cdot b_i - \sum_{e \in E(i)} \frac{x_{i,e}^2}{2}.$$

For fixed π and fixed x_{-i} (the investments of other defenders), V_i is a sum of two terms in x_i. The first term is linear in each $x_{i,e}$ (the survival product is multilinear in the $x_{j,e}$), hence concave. The second term is strictly concave in each $x_{i,e}$ as a quadratic with negative leading coefficient. The sum is concave in x_i, hence quasi-concave.

All conditions of the Debreu-Fan-Glicksberg theorem are satisfied. A Nash equilibrium exists. □

A.2 Pure Nash Equilibrium

[Pure NE conditions] A pure Nash equilibrium exists if and only if there is a target i^* such that

(i) $b_{i^*}(1 - b_{i^*}) \geq b_j$ for every target $j \neq i^*$ such that some path from 0 to j does not go through i^*;

(ii) $b_{i^*} \geq b_j$ for every target $j \neq i^*$ such that every path from 0 to j goes through i^*.

Strategy. In a pure equilibrium the attacker concentrates on a single target along a single path, and the defender of that target invests on the single edge through which the attack arrives. The model collapses to its node-based form on the relevant fragment, and the conditions are unchanged from Bloch's analysis.

Proof. Suppose the attacker plays a pure strategy concentrating on target i^* via path p^*, and the only investment used in equilibrium is $x_{i^*,e^*} = b_{i^*}$ on the final edge $e^* \in p^*$. The attacker's payoff is $b_{i^*}(1 - b_{i^*})$.

The attacker has no profitable deviation to a target j off the path through i^*: such a deviation yields payoff b_j (against zero defense), so we need $b_{i^*}(1 - b_{i^*}) \geq b_j$. This is condition (i).

For a target j behind i^* (every path goes through i^*), deviating to j requires surviving i^*'s defense, which gives expected payoff $b_j(1 - b_{i^*})$. We need $b_{i^*}(1 - b_{i^*}) \geq b_j(1 - b_{i^*})$, that is, $b_{i^*} \geq b_j$. This is condition (ii).

Conversely, if both conditions hold, the strategy profile described is a best response on each side: the attacker's payoff is maximized at i^*, and the defender's investment b_{i^*} on e^* is the unique solution to the first-order condition when $q_{i^*} = 1$. □

A.3 Lemmas Carrying Over from Bloch

A.3.1 Lemma 1: Equal Payoff Across the Support

In equilibrium, all targets i in the attacker's support yield the same expected payoff U.
Strategy. If two targets in the support gave different expected payoffs, the attacker would shift weight from the lower-payoff target to the higher-payoff one, contradicting equilibrium. The argument depends only on the attacker's optimization and is independent of where defense is placed.

Proof. Suppose two targets i and j in the support give different expected payoffs, say target i gives $V_i^{\text{att}} > V_j^{\text{att}}$. The attacker can increase utility by transferring mass from any path leading to j onto any path leading to i, which contradicts the assumption that the current π is a best response. Hence all targets in the support yield the same payoff U. □

A.3.2 Lemma 3: Optimal Predecessor Has Lowest Value

If a target i is reachable from the attacker through multiple support members, the attacker's optimal route to i goes through the support member with the lowest value.
Strategy. A predecessor with lower value invests less in equilibrium (since investments scale with target value), so survival through that predecessor is higher. The attacker maximizes survival probability by choosing the lowest-value predecessor.

Proof. Let j_1, j_2 be two support members through which i can be reached, with $b_{j_1} < b_{j_2}$. In equilibrium, the investment of node j on the edge leading toward i is $1 - U/b_j$ for first targets in their respective support positions (or $1 - b_k/b_j$ for deeper targets). In either case, the survival factor through j increases as b_j decreases, because the investment $1 - U/b_j$ decreases as b_j decreases.

Hence routing through j_1 gives higher path survival than routing through j_2. The attacker prefers the route through j_1. □

A.3.3 Lemma 4: Values Increase Along Attack Paths

Let $i_1, i_2, \ldots, i_k$ be the support members on a path from 0 to a target i_k, in order. Then $b_{i_1} < b_{i_2} < \cdots < b_{i_k}$.
Strategy. If a deeper target on a path had a lower value than an earlier one, attacking the deeper target would be strictly worse than attacking the earlier one, so the attacker would not include the deeper target in the support.

Proof. Consider two consecutive support members on the path, i_ℓ and $i_{\ell+1}$. The attacker's expected payoff from attacking $i_{\ell+1}$ via this path is $\alpha_{i_{\ell+1}} \cdot b_{i_{\ell+1}}$, where $\alpha_{i_{\ell+1}}$ is the survival to $i_{\ell+1}$. The attacker's expected payoff from attacking i_ℓ via the same path (truncated at i_ℓ) is $\alpha_{i_\ell} \cdot b_{i_\ell}$.

By Lemma A.3.1, both equal U. We have $\alpha_{i_{\ell+1}} < \alpha_{i_\ell}$ (since the attack must survive additional defense beyond i_ℓ). Thus $b_{i_{\ell+1}} > b_{i_\ell}$. The values increase strictly along the path. □

A.4 Modified Lemma 2: Mutual Coverage on Free-Highway Edges

If a first target i in the support has m_i free-highway incoming edges, then in equilibrium the attacker mixes uniformly across all m_i such edges, and the defender invests an equal amount on each.

Strategy. If the defender concentrated investment on one of the incoming edges and the attacker concentrated attack on that same edge, then any other free-highway incoming edge would be undefended, and the attacker would have a strictly profitable deviation to that edge. The only consistent outcome is for both sides to spread effort across all m_i edges symmetrically.

Proof. Suppose for contradiction that the attacker assigned a positive probability q to one free-highway edge e_1 to target i and a strictly lower probability $q' < q$ to another free-highway edge e_2 to the same target i.

The defender's first-order condition on each edge equates marginal benefit (the attack probability through that edge times the value of the target) to marginal cost (proportional to the investment). Lower attack probability on e_2 means lower investment $x_{i,e_2} < x_{i,e_1}$. But then the attacker's expected payoff from using e_2 is $b_i(1 - x_{i,e_2}) > b_i(1 - x_{i,e_1})$, contradicting the assumption that e_1 is part of an optimal mixed strategy.

The same argument rules out asymmetric defender investments. The only consistent outcome is uniform mixing by the attacker and equal investment by the defender on all m_i free-highway edges.

This is the continuous analog of the mutual coverage condition in Mavronicolas et al. (2008), where defender and attacker coverage match at the level of individual edges rather than nodes. □

Important note on m_i. The factor m_i counts *distinct edges incident to the target*, not paths from the attacker. Multiple paths converging to the same final edge before reaching i contribute $m_i = 1$, not more. This distinction matters in networks where the attacker has many alternative routes that share their last edge into the target.

A.5 First-Target Equilibrium Equations

[First-target equilibrium] For a first target i in the support, with m_i free-highway incoming edges:

$$x_{i,e} = 1 - \frac{U}{b_i}, \qquad q_i = \frac{m_i(b_i - U)}{b_i^2}.$$

The investment is the same on each of the m_i free-highway edges.

Strategy. The defender's first-order condition on a single free-highway edge gives the per-edge investment as a function of the attack probability through that edge. Combining with the attacker's indifference condition (Lemma A.3.1) gives the per-edge investment in terms of U, which then determines the total attack probability q_i once we sum over all m_i edges.

Proof. Let e be a free-highway incoming edge to target i. By Lemma A.4, the attacker's mass on e is q_i/m_i, and the defender invests an equal amount on each such edge.

The defender's first-order condition for $x_{i,e}$ requires

$$\frac{q_i}{m_i} \cdot b_i = x_{i,e},$$

where the left-hand side is the marginal benefit (probability of attack through this edge times the avoided loss) and the right-hand side is the marginal cost.

The attacker's indifference condition (Lemma A.3.1) yields

$$b_i(1 - x_{i,e}) = U \quad \Longrightarrow \quad x_{i,e} = 1 - \frac{U}{b_i}.$$

Substituting into the defender's first-order condition:

$$\frac{q_i}{m_i} \cdot b_i = 1 - \frac{U}{b_i} \quad \Longrightarrow \quad q_i = \frac{m_i(b_i - U)}{b_i^2}.$$

The feasibility condition $U/b_i < 1$ ensures $q_i > 0$ for every supported first target. □

A.6 Deeper-Target Equilibrium Equations

[Deeper-target equilibrium] For a deeper target i in the support with immediate predecessor k:

$$x_{i,e_i} = 1 - \frac{b_k}{b_i}, \qquad q_i = \frac{b_k(b_i - b_k)}{U \cdot b_i^2},$$

where e_i is the edge connecting k to i on the attack path.

Strategy. A deeper target has a single relevant incoming edge from its predecessor in the support. The proof reduces to the Bloch case because there is only one edge to defend and one mass of attack probability to consider.

Proof. The attacker's indifference between attacking k and i along the same path gives

$$\alpha_k \cdot b_k = \alpha_k(1 - x_{i,e_i}) \cdot b_i,$$

where α_k is the common path survival up to (but not through) the edge e_i. Cancelling α_k:

$$1 - x_{i,e_i} = \frac{b_k}{b_i} \quad \Longrightarrow \quad x_{i,e_i} = 1 - \frac{b_k}{b_i}.$$

The defender's first-order condition gives $q_i \cdot b_i = x_{i,e_i}$ via the route through the predecessor's edge structure (the attacker's probability of routing to i through the path of interest is q_i). However, the analysis must be conducted on the survival-weighted attack probability. Substituting and using $\alpha_k \cdot b_k = U$:

$$q_i = \frac{b_k(b_i - b_k)}{U \cdot b_i^2}.$$

□

A.7 Uniqueness of Equilibrium

[Uniqueness] Under the genericity assumption ($b_i \neq b_j$ for all $i \neq j$), the edge-based defense game has a unique Nash equilibrium.

Strategy. The argument follows the same structure as Bloch et al. (2023, Theorem 1), with survival probabilities replaced by their edge-based analogs. The key idea is to suppose two equilibria with different supports Δ_1 and Δ_2 exist, derive a contradiction via the attacker's best-response condition, and conclude that the supports must coincide. Once the supports coincide, the equilibrium equations from Propositions A.5 and A.6 pin down all remaining quantities.

Proof. Suppose for contradiction that two distinct equilibria exist, with attacker supports Δ_1 and Δ_2 and corresponding payoffs U_1 and U_2. By the genericity assumption, $U_1 \neq U_2$ generically; without loss of generality, $U_1 < U_2$.

Step 1: The supports cannot coincide. If $\Delta_1 = \Delta_2$, then the equilibrium equations (Propositions A.5 and A.6) determine U uniquely as the root of $\sum_{j \in \Delta} q_j(U) = 1$. Since each q_j is strictly decreasing in U (verified in the comparative-statics derivation), the sum is strictly decreasing, so the equation has at most one root. This contradicts $U_1 \neq U_2$. Hence $\Delta_1 \neq \Delta_2$.

Step 2: One support is contained in the other. Suppose neither support contains the other; then there exist $i \in \Delta_1 \setminus \Delta_2$ and $j \in \Delta_2 \setminus \Delta_1$. By the construction of equilibrium support, target i in equilibrium 1 satisfies $b_i(1 - x_{i,e}^{(1)}) = U_1$ on its incoming edge, while in equilibrium 2 it has zero defense (since $i \notin \Delta_2$), giving deviation payoff b_i. Combined with $U_2 > U_1$, the only way for $i \notin \Delta_2$ to be consistent with equilibrium 2 is $b_i \leq U_2$. By a symmetric argument on j: $b_j \leq U_1 < U_2$, but $j \in \Delta_2$ requires $b_j > U_2$, a contradiction. Hence one support strictly contains the other; without loss of generality $\Delta_2 \subset \Delta_1$.

Step 3: The smaller support gives a higher payoff. With $\Delta_2 \subset \Delta_1$ strictly, the support has fewer members in equilibrium 2, so the constraint $\sum_{j \in \Delta_2} q_j(U_2) = 1$ involves fewer (positive) terms. Each remaining q_j must therefore be larger on average, which (since each q_j decreases in U) requires $U_2 > U_1$, consistent with our assumption.

Step 4: The smaller-support equilibrium is not a best response. Take any target $i \in \Delta_1 \setminus \Delta_2$. By Step 2, $b_i \leq U_2$. But i is in equilibrium 1's support, which by the equilibrium equations requires $b_i > U_1$. From $U_1 < U_2$ and $b_i \leq U_2$, we have $U_1 < b_i \leq U_2$. In equilibrium 2, target i has zero defense (it is excluded), so the attacker's deviation payoff to i is exactly b_i. The attacker's equilibrium payoff in equilibrium 2 is $U_2 \geq b_i$ *only when equality holds*, but generically $U_2 > b_i$ would mean i should be in Δ_2 (contradicting its exclusion), and $U_2 = b_i$ violates genericity. Either case yields a contradiction.

The supports must therefore coincide, and by Step 1 the equilibrium is unique. □

A.8 Strategic Substitutes on Shared Edges

[Strategic substitutes] On a shared edge $e = (i, j)$ where both endpoints have incentive to invest, the defenders' best responses are strategic substitutes:

$$\frac{\partial x_{i,e}^*}{\partial x_{j,e}} < 0.$$

Strategy. Differentiate defender i's first-order condition on the shared edge with respect to defender j's investment. The presence of the $(1 - x_{j,e})$ factor in the survival product produces a strictly negative cross-partial.

Proof. Let q be the attack probability flowing through edge e targeting node i. Defender i's first-order condition on e requires

$$x_{i,e}^* = q \cdot b_i \cdot (1 - x_{j,e}).$$

Differentiating with respect to $x_{j,e}$:

$$\frac{\partial x_{i,e}^*}{\partial x_{j,e}} = -q \cdot b_i.$$

Since $q > 0$ (defender i has positive incentive to invest) and $b_i > 0$, the derivative is strictly negative. □

A.8.1 Free-Riding Cannot Drive Investments to Zero

In any equilibrium, the pair $(x_{i,e}, x_{j,e}) = (0, 0)$ cannot be part of an equilibrium on an edge $e = (i, j)$ where both endpoints have positive attack probability flowing through.

Strategy. If one endpoint invested zero, the other's best response would be strictly positive (proportional to the attack probability and target value), contradicting the assumption that both invest zero.

Proof. Suppose $x_{j,e} = 0$. Then defender i's first-order condition gives $x^*_{i,e} = q \cdot b_i \cdot 1 = q \cdot b_i > 0$. So $(x_{i,e}, x_{j,e}) = (0, 0)$ is not consistent with both defenders' best responses simultaneously, and cannot be part of any equilibrium. □

A.9 Comparative Statics

[Effect of changing target value] The equilibrium attacker utility U changes monotonically with the value b_i of a supported target i, with the direction depending on whether b_i is above or below $2U$.

Strategy. Apply implicit differentiation to the equilibrium constraint $\sum_{j\in\Delta} q_j(U; b_j) = 1$. The sign of $\partial U/\partial b_i$ depends on $2U - b_i$, which determines whether raising b_i benefits or harms the attacker.

Proof. Take the equilibrium constraint $\sum_{j\in\Delta} q_j = 1$ and differentiate implicitly with respect to b_i:

$$\frac{\partial q_i}{\partial b_i} + \frac{\partial q_i}{\partial U} \cdot \frac{\partial U}{\partial b_i} + \sum_{j\neq i} \frac{\partial q_j}{\partial U} \cdot \frac{\partial U}{\partial b_i} = 0.$$

For a first target, $q_i = m_i(b_i - U)/b_i^2$, so

$$\frac{\partial q_i}{\partial b_i} = \frac{m_i(2U - b_i)}{b_i^3}, \quad \frac{\partial q_i}{\partial U} = -\frac{m_i}{b_i^2} < 0.$$

For all $j \neq i$ with j a first target, $\partial q_j/\partial U = -m_j/b_j^2 < 0$. For a deeper target j with predecessor k, $q_j = b_k(b_j - b_k)/(Ub_j^2)$, so $\partial q_j/\partial U = -b_k(b_j - b_k)/(U^2 b_j^2) < 0$.

Solving for the comparative static:

$$\frac{\partial U}{\partial b_i} = \frac{\partial q_i/\partial b_i}{|\sum_{j\in\Delta} \partial q_j/\partial U|} = \frac{m_i(2U - b_i)/b_i^3}{\sum_j m_j/b_j^2 + (\text{deeper-target terms})}.$$

The denominator is strictly positive. The sign of the derivative therefore matches the sign of $2U - b_i$.

Two regimes arise. When $U > b_i/2$, the numerator is positive and an increase in b_i raises U. This is the typical regime for targets close to dropping out of the support, where the equilibrium U is close to b_i. When $U < b_i/2$, the numerator is negative and an increase in b_i actually lowers U. This second regime arises for targets whose value is substantially above the equilibrium attacker payoff. Intuitively, raising the value of a target that the attacker is already strongly attracted to does not change behavior much, but raising the value of a target on the margin does.

In the typical case — where the support is determined by targets whose values are not far above U — the first regime applies and the derivative is positive. $\square$

[Adding a connection] Adding a free-highway incoming edge to a supported first target i raises m_i by one and strictly increases the attacker's equilibrium utility U.
Strategy. The new edge enters the equilibrium constraint through m_i, increasing the contribution of q_i to the sum. Implicit differentiation shows that the resulting U must rise.

Proof. With m_i replaced by $m_i + 1$, the constraint $\sum_{j\in\Delta} q_j = 1$ is no longer satisfied at the original U:

$$\sum_{j\neq i} q_j(U) + (m_i + 1)(b_i - U)/b_i^2 > 1.$$

For the constraint to hold, U must rise, since each q_j is a decreasing function of U. The change is strict: $\partial U/\partial m_i > 0$.

Equivalently, by implicit differentiation:

$$\frac{\partial U}{\partial m_i} = \frac{(b_i - U)/b_i^2}{|\sum_{j\in\Delta} \partial q_j/\partial U|} > 0.$$

$\square$

This rules out the Braess-type paradox that can arise in Bloch's node-based model. Adding a connection to a first target unambiguously benefits the attacker in the edge-based formulation.

[Eliminating a target] If a target i is removed from the attacker's consideration (forced out of the support), the equilibrium utility U for the remaining targets weakly increases if and only if i was the lowest-contribution member of the support.
Strategy. Removing i subtracts q_i from the sum. The remaining members' attack probabilities adjust so the new sum equals one. If i contributed less than the average, removing it requires a higher U to make up the deficit.

Proof. Let $\Delta' = \Delta \setminus \{i\}$. The new equilibrium constraint is $\sum_{j\in\Delta'} q_j(U') = 1$. Since each q_j is decreasing in U, and the sum is now over fewer terms, the new U' must satisfy $U' \geq U$ if and only if $q_i(U) <$ the "average contribution" needed for the remaining members, which is the case when i's contribution q_i was small relative to the others. $\square$

A.10 Cooperative Defense: Symmetry Within Edges

[Symmetry within edges] In the cooperative equilibrium, the two endpoints of any edge $e = (i, j)$ on which both have positive incentive to invest contribute equally: $y_{i,e} = y_{j,e}$.
Strategy. The cooperative loss function is symmetric in the two investments on a shared edge — both enter as $(1 - y)$ in the survival product, and both have identical quadratic cost terms. The first-order conditions are therefore symmetric, and the unique solution sets them equal.

Proof. The cooperative defender minimizes total expected loss subject to the equilibrium structure. The contribution of edge $e = (i, j)$ to the loss is

$$\sum_k r_k \cdot (\text{survival up to } e) \cdot (1 - y_{i,e})(1 - y_{j,e}) \cdot (\text{survival after } e) \cdot b_k + \frac{y_{i,e}^2}{2} + \frac{y_{j,e}^2}{2},$$

where the sum runs over targets k whose attack path passes through e.

The first-order condition for $y_{i,e}$ is

$$y_{i,e} = (1 - y_{j,e}) \cdot \sum_k r_k \cdot \alpha_e \cdot b_k,$$

where α_e is shorthand for the survival-product factors not involving $y_{i,e}$ or $y_{j,e}$. The first-order condition for $y_{j,e}$ has identical form with i and j swapped.

Subtracting the two conditions:

$$y_{i,e} - y_{j,e} = (1 - y_{j,e}) \cdot S - (1 - y_{i,e}) \cdot S = S(y_{i,e} - y_{j,e}),$$

where $S = \sum_k r_k \cdot \alpha_e \cdot b_k > 0$.

Rearranging: $(1 - S)(y_{i,e} - y_{j,e}) = 0$. If $S = 1$, the equation is trivially satisfied for any pair, but then the system has a degenerate one-parameter family of solutions; the unique pinned-down solution is the symmetric one $y_{i,e} = y_{j,e}$. If $S \neq 1$, then $y_{i,e} = y_{j,e}$ directly. □

A.11 Cooperative Cost Savings on Lines

[Cost savings on lines] On a line network, the cooperative edge-based defense achieves the same equilibrium attacker utility U as the cooperative node-based (Bloch) defense, but with strictly lower total defense cost.
Strategy. The attacker's indifference condition gives the same effective interception per position in both models, hence the same U. The cost saving comes from the convexity of the quadratic cost: splitting an investment y between two endpoints, each contributing $y/2$, costs $2(y/2)^2/2 = y^2/4$, strictly less than the single-node cost of $y^2/2$.

Proof. Consider a line network with support Δ and node-based cooperative equilibrium investments y_p^N at position p, $p = 1, \ldots, n$. Define the edge-based cooperative investments y_p^E as the symmetric solution at each interior edge such that $(1 - y_p^E)^2 = (1 - y_p^N)$. This gives

$$y_p^E = 1 - \sqrt{1 - y_p^N}.$$

By construction, the survival probability at each position is the same in both models, so the attacker's indifference conditions are satisfied with the same U.

The cost at position p in the node model is $(y_p^N)^2/2$. The cost at position p in the edge model (interior position with two investors) is $2 \cdot (y_p^E)^2/2 = (y_p^E)^2$.

We compare $(y_p^N)^2/2$ to $(y_p^E)^2$. Using the change of variables $u = 1 - y_p^N$, so $y_p^N = 1 - u$ and $y_p^E = 1 - \sqrt{u}$ (with $0 < u < 1$):

$$\begin{aligned}
\frac{(y_p^N)^2}{2} - (y_p^E)^2 &= \frac{(1-u)^2}{2} - (1 - \sqrt{u})^2 \\
&= \frac{1 - 2u + u^2}{2} - (1 - 2\sqrt{u} + u) \\
&= \frac{1}{2} - u + \frac{u^2}{2} - 1 + 2\sqrt{u} - u \\
&= -\frac{1}{2} - 2u + \frac{u^2}{2} + 2\sqrt{u}.
\end{aligned}$$

For $0 < u < 1$, we can verify numerically (or by analyzing the function) that this expression is strictly positive. For instance at $u = 0.5$: $-0.5 - 1 + 0.125 + \sqrt{2} \approx -1.375 + 1.414 > 0$.

The entrance edge (position 1) has only one investor in the edge model, so its cost is identical to the node model. The cost saving therefore comes from the interior positions, and is strictly positive. $\square$

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